

Trainings ▾

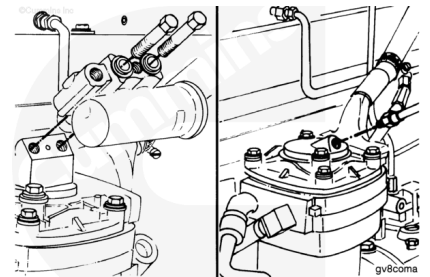
Initial Check

⚠ WARNING ⚠

Air pressure must be released from system before removing the air governor to reduce the possibility of personal injury.

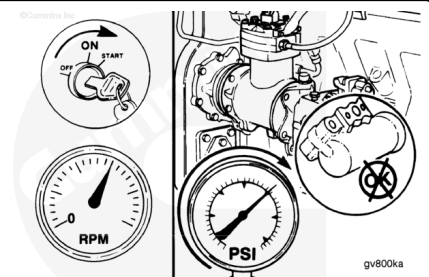
Note : The illustrations shown will be of the SS model single-cylinder air compressor. Differences in procedures for SS, QE, and ST model Holset® air compressors will be shown where necessary.

Remove the air governor or air governor hose from the air compressor unloader body.



Operate the engine to activate the air compressor.

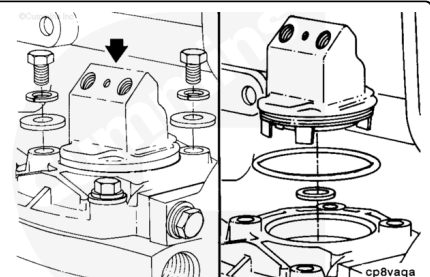
If the air compressor is **not** pumping, the unloader valve is malfunctioning and **must** be repaired or replaced.



Disassemble

⚠ WARNING ⚠

The unloader valve body is installed with spring tension. To reduce the possibility of personal injury, always wear protective eye wear and use care when removing the unloader valve body.



The following steps are for the Holset® SS, E-Type, and ST models.

Hold the unloader valve body down, and remove the two captive washer capscrews and the two plain washers.

Remove the unloader valve body.

Remove the o-ring seal.

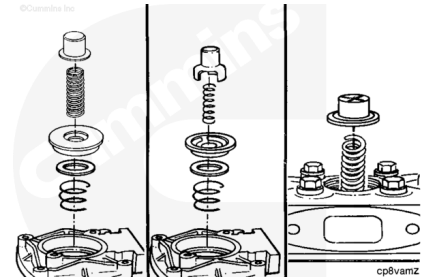
Remove the rectangular ring seal.

Note : Disassembly of the center unloader valve on Holset® two-cylinder air compressors is similar to the single-cylinder unloader valve.

Remove the unloader valve cap and the unloader valve spring.

Remove the intake valve seat and valve.

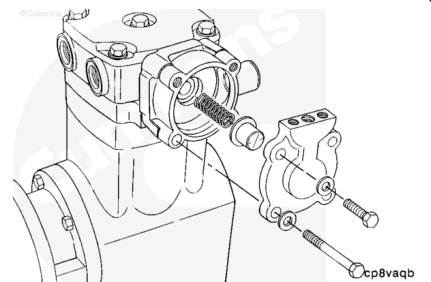
Remove the intake valve spring.



Holset® QE230, QE296, and QE338 A/C Model

⚠ WARNING ⚠

The unloader valve body is installed with spring tension. To reduce the possibility of personal injury, always wear protective eye wear and use care when removing the unloader valve body.



Hold the unloader valve body down and remove the capscrews.

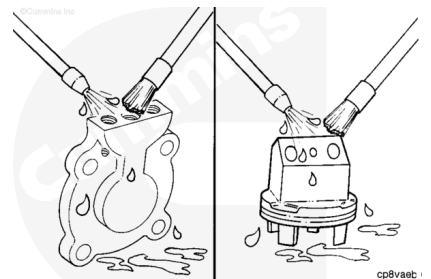
Remove the unloader body, unloader cap and spring.

Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not use caustic cleaners. Caustic cleaners can cause damage to component parts.

Remove all carbon and varnish from the unloader valve cap body.



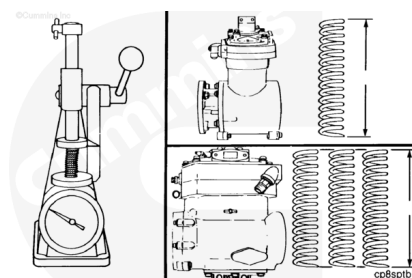
For Holset® two-cylinder air compressors, check both cylinder and center unloader springs. Holset® Engineering Co., Inc. recommends that new springs be installed during rebuild.

Use a valve spring tester to check the unloader spring.

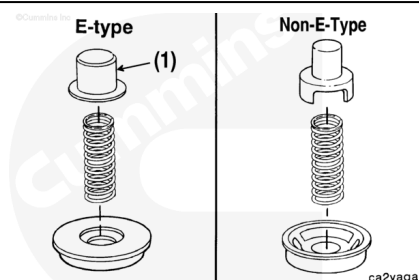
Refer to Procedure 018-022

(/qs3/pubsys2/xml/en/procedures/04/04-018-022.html) for the Compressor Spring Force Specifications chart.

Replace the unloader spring if it does **not** meet the specifications shown, or the wrong spring has been used.



If the compressor has a flat hat type of unloader cap (1), it **must** use an unloader spring and valve seat different from that used with the three-prong unloader.



Assemble

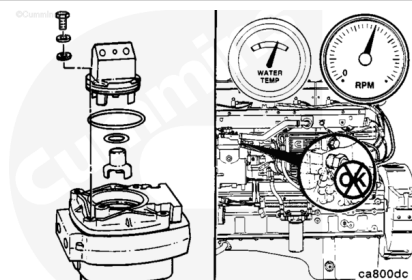
The following steps are for the Holset® SS, E-Type, and ST models.

Apply high-temperature Accrolube lubrication teflon grease, or equivalent, to the rectangular ring seal, unloader cap, and unloader body bore.

Install the unloader spring, cap and valve body.

Tighten the capscrews.

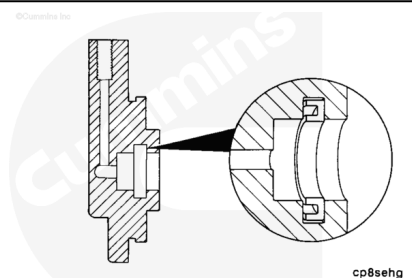
Torque Value: 14 n•m [120 in-lb]



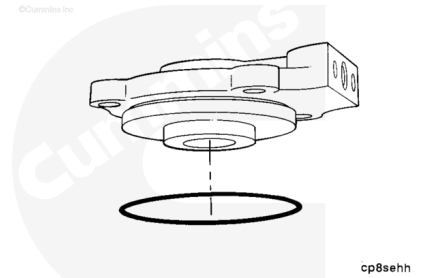
Holset® QE230, QE296, and QE338 A/C Model

Install the new rectangular V-seal into the unloader body with the grooved side up.

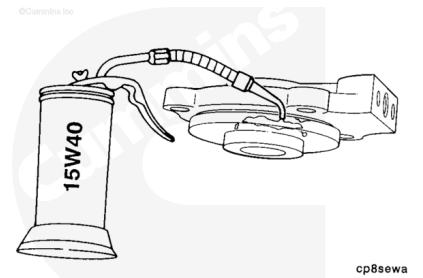
Liberally lubricate the unloader valve bore above and below the rectangular ring seal with high-temperature Accrolube lubrication Teflon grease, or equivalent.



Install a new o-ring seal on the unloader valve body.



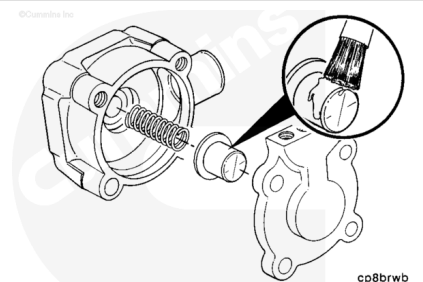
Use clean 15W-40 oil or Accrolube lubrication Teflon grease, or equivalent, to lubricate the seal.



Liberaly lubricate the unloader valve body bore and unloader cap with high-temperature Accrolube lubrication Teflon grease, or equivalent.

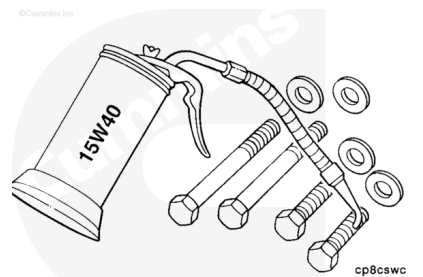
Install the unloader cap.

Install the unloader spring.



Note : The two unloader body screws must not be used to attach any brackets.

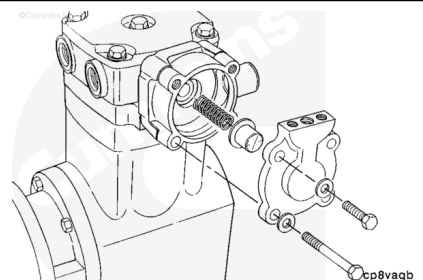
Lubricate the unloader capscrew threads and under head with clean engine oil (SAE 15W-40) before installation.



The longer capscrews are used to mount the manifold to the air compressor.

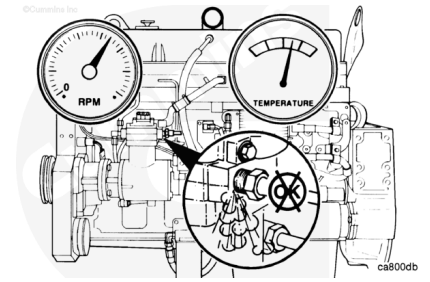
Assemble the unloader components, and attach the unloader assembly to the valve plate with the four capscrews and washers.

Torque Value: 27 n•m [20 ft-lb]



Finishing Steps

Operate the engine and check for leaks.



Last Modified: 13-Jun-2005
